

# PENGZHEN LI

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## EDUCATION

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|-----------------------------------------------------------------|---------------|
| <b>University of Tennessee, Knoxville</b><br>Ph.D. in Economics | expected 2027 |
| <b>Shandong University</b><br>M.A. in Finance                   | 2018          |
| <b>Tongji University, Shanghai</b><br>B.A. in Economics         | 2015          |

## RESEARCH INTERESTS

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International Trade, Trade Policy, Applied Econometrics, and Political Economy.

## WORKING PAPERS

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**Trade Restrictiveness Indices with Discriminatory Tariffs** with James Lake [**Job Market Paper**]. *Presented at the Midwest Economic Theory and International Trade Meetings, Virginia Tech, 2025.*

Abstract: Trade-restrictiveness indices summarize a country's tariff schedule as a single welfare-equivalent uniform tariff, but the existing indices assume most-favored-nation tariffs. Since 2018 the largest tariff changes—the US–China tariff war, retaliatory tariffs, and proliferating preferential agreements—have been discriminatory by design. We extend the trade-restrictiveness index to partner-specific tariffs while preserving its welfare interpretation and derive a closed-form decomposition into a “level” component (how much a country restricts trade) and a “discrimination” component (how unevenly it does so). Computing the indices from WITS/Comtrade data for 2017–2024, we find that the MFN-only index systematically understates trade distortions when discrimination is high, and the decomposition separates, for the first time in this framework, the welfare cost of discrimination from the cost of protection.

**The Impact of Mayoral Endorsements on Presidential Election Outcomes** with Matt Van Essen and Luiz Lima. *2024 J. Fred and Wilma Holly Award (best second-year paper).*

Abstract: We develop and estimate a structural model of mayoral endorsements in presidential elections as an anonymous Bayesian game with mean-field equilibrium. The mean-field formulation reduces equilibrium computation to a fixed point on the candidate simplex, independent of the number of mayors, making structural estimation tractable at the scale of an entire country. Identification follows a BLP-style logit inversion of observed log-odds onto type covariates and aggregate shares. We apply the framework to Brazil's coalition-mediated endorsement system (2014, 5,623 mayors) and France's individual parrainage system (2017 and 2022).

**Price Risk, Irreversible Investment, and the Welfare Benchmark for the Section 45Z Clean Fuel Production Credit** with T. Edward Yu.

**How Should 45Z Pay for Risk? A Welfare-CVaR and Project-Finance Decomposition of the IRA Clean Fuel Production Credit** with T. Edward Yu.

**Stochastic Optimisation of Logging Residue-based Biofuel Supply Chains Considering Feedstock Ash Content and Biofuel Price** with T. Yu et al..

**Sustainable Aviation Fuel Production from Forest Residues with Ash Content: A Case**

**Study of Nashville Airport** with T. Yu et al..

**Estimating Pork Primal Price Relationships Using a Threshold Vector Autoregressive Analysis** with Charles Martinez, Christopher N. Boyer, T. Edward Yu, and David Anderson. *SSRN working paper 4121953* (2022).

## WORK IN PROGRESS

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**Cooperative Multilateral Tariff Negotiations: A Quantitative Analysis Using Shapley Value** with Georg Schaur. *Draft v2, August 2026*.

## PEER-REVIEWED PUBLICATIONS

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“Spatial optimization of the sustainable aviation fuel supply chains from forest residues via fast pyrolysis/hydrotreatment considering feedstock ash content variability.” **P. Li**, T. E. Yu, N. Labbé, N. Abdoulmoumine, M. Garcia-Perez, K. P. Hoyt, and B. C. English. *Biomass and Bioenergy*, 208 (2026), 108793.

“Assessing the impact of preprocessing and conversion technologies on the sustainable aviation fuel supply from forest residues in the Southeast USA.” **P. Li**, T. E. Yu, C. Trejo-Pech, J. A. Larson, B. C. English, and D. N. Lanning. *Transportation Research Record*, 2678(9) (2024), 639–654.

“Beef price spread relationship with processing capacity utilization.” C. Martinez, **P. Li**, C. N. Boyer, T. E. Yu, and J. G. Maples. *Journal of the Agricultural and Applied Economics Association*, 2(1) (2023), 133–145.

“Do green technology innovations contribute to carbon dioxide emission reduction? Empirical evidence from patent data.” K. Du, **P. Li**, and Z. Yan. *Technological Forecasting and Social Change*, 146 (2019), 297–303.

## PRESENTATIONS

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|--------------------------------------------------------------------------------------|-------------|
| <b>Midwest Economic Theory and International Trade Meetings</b> , Virginia Tech      | Spring 2025 |
| <b>Western Regional Science Association</b> , Annual Meeting, Hawaii                 | 2023        |
| <b>Western Agricultural Economics Association</b> , Annual Meeting, Santa Fe, NM     | 2022        |
| <b>Southern Agricultural Economics Association</b> , Annual Meeting, New Orleans, LA | 2022        |

## TEACHING EXPERIENCE

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### University of Tennessee

#### *Instructor*

|                                                     |                          |
|-----------------------------------------------------|--------------------------|
| ECON 351: Monetary Economics                        | Spring 2025, Spring 2026 |
| ECON 322: The Global Economy: Trade and Development | Summer 2025              |
| ECON 313: Intermediate Macroeconomics               | Summer 2026              |

#### *Teaching Assistant*

##### Ph.D. courses:

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|---------------------------------------------|---------------------------------|
| ECON 511: Microeconomic Theory              | Fall 2023, Fall 2024, Fall 2025 |
| ECON 512: Advanced Microeconomics           | Spring 2024                     |
| ECON 581: Mathematical Methods in Economics | Summer 2024                     |

##### Master's courses:

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| AREC 525: Agribusiness Operations Research Methods | Spring 2022, Spring 2023 |
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## PROFESSIONAL EXPERIENCE

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### **University of Tennessee – Knoxville, TN**

2020 – present

Graduate Research & Teaching Assistant. USDA NIFA and FAA ASCENT projects (2020–2023): mixed-integer supply-chain optimization (GAMS), IMPLAN/DEA siting analysis, and spatial data pipelines (ArcGIS, R, Python) for forest-residue sustainable aviation fuel; two publications and three working papers.

### **Shandong University – Jinan, China**

2016 – 2018

Graduate Research Assistant. Panel-threshold and machine-learning analysis of green innovation and CO<sub>2</sub> emissions (published in *TFSC*, 2019).

## HONORS AND AWARDS

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|---------------------------------------------------------------------------------|----------------|
| Charles B. Garrison Award for Excellence in Teaching, University of Tennessee   | 2026           |
| Graduate Summer Fellowship, University of Tennessee                             | 2025           |
| J. Fred and Wilma Holly Award (best second-year paper), University of Tennessee | 2024           |
| J. Fred and Wilma Holly Fellowship, University of Tennessee                     | 2023 – present |
| Graduate Assistantship, University of Tennessee                                 | 2020 – present |
| Outstanding Volunteer, migrant-family tutoring program, Shanghai                | pre-Ph.D.      |

## SKILLS

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**Programming Languages:** Stata, R, Python, SQL, GAMS, MATLAB, LaTeX, ArcGIS, IMPLAN.

**Methods:** panel and causal econometrics (FE, IV, DiD, event study, SCM, GMM, panel threshold), time series (TVAR, VECM, mGARCH), mixed-integer and stochastic programming, DEA, cooperative and mean-field games, machine learning.

**Credentials:** CFA Level II.

**Languages:** Chinese (native), English (fluent).

## REFERENCES

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### James Lake

Associate Professor of Economics

Department of Economics

University of Tennessee

Email: [jlake7@utk.edu](mailto:jlake7@utk.edu)

### Georg Schaur

Leigh Burch III Economics Professor and Haslam Family Faculty Fellow

Department of Economics

University of Tennessee

Email: [gschaur@utk.edu](mailto:gschaur@utk.edu)

### Matthew Van Essen

Lena M. and Clarence A. Garvin, Sr. Professor and Dr. Joe Johnson Faculty Fellow

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Email: [mvanesse@utk.edu](mailto:mvanesse@utk.edu)